Modelling and Vibration Analysis of a Crane Robot Using MATLAB

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MODULE NO	5FTC2145
DATE	12.03.2025

Contents

01. Abstract 4
02. Introduction 4
03. Task 01 analysis 4
1) Robot Topology 4
2) Equivalent Mass Spring Damper Model 6
3) Vibration Problems in the Crane System 6
4) Vibration Control Methods 7
5) Similar Robotic Systems with Vibration Problems 8
04. Task 02 MATLAB results 8
1) Data Selection and Plotting 8
2) Parameter Identification 9
3) Peak Detection 9
4) Logarithmic Decrement 10
5) Damping Ratio 11
6) Damped Natural Frequency 11
7) Natural Frequency 11
8) Rope Length 12
9) Model Verification 12
10) Identified System Parameters 13
05. Discussion 13
06. Conclusion 14
07. References 14
08. Appendix 15
MATLAB code used for system identification 15
GITHUB REPO 17

Figure 1 Model of a crane 5
Figure 2 how different damping works in a crane 8
Figure 3 Vibration signal of the crane pendulum system 9

01. Abstract

This report presents the work completed for Coursework for Mechatronics Module . Task 01 analyses a simplified robot crane system, including its joint topology, mechanical structure, vibration behavior, and possible vibration control methods. Similar robotic systems that experience comparable vibration issues are also discussed. Task 02 uses MATLAB to analyses vibration data . The logarithmic decrement method is applied to estimate the damping ratio Early research on command shaping for vibration suppression was introduced by Singer and Seering[5], natural frequency, and rope length of the crane's pendulum system.

02. Introduction

Robot crane systems are widely used in industrial environments such as seaports, warehouses, and construction sites to transport heavy loads efficiently[1]. These systems consist of mechanical and control components that work together, making them an important example of a mechatronic system. During operation, the suspended load connected by a cable can swing and create vibrations, especially when the crane moves quickly or experiences external disturbances. These vibrations can reduce positioning accuracy, increase operation time, and affect the overall efficiency of the robotic system.

Understanding and modelling the vibration behavior of crane systems is important for improving performance and safety. The motion of the suspended load can be approximated as a pendulum, and the system dynamics can be represented using a second-order model similar to a mass spring damper system. Analyzing these dynamics helps engineers design better control methods to reduce unwanted oscillations.

03. Task 01 analysis

1) Robot Topology

The simplified crane robot shown in the coursework consists of a vertical mast, a horizontal beam, a trolley system, a suspension cable, and a load hanging at the end. Each connection in the system can be described using its mechanical degree of freedom.

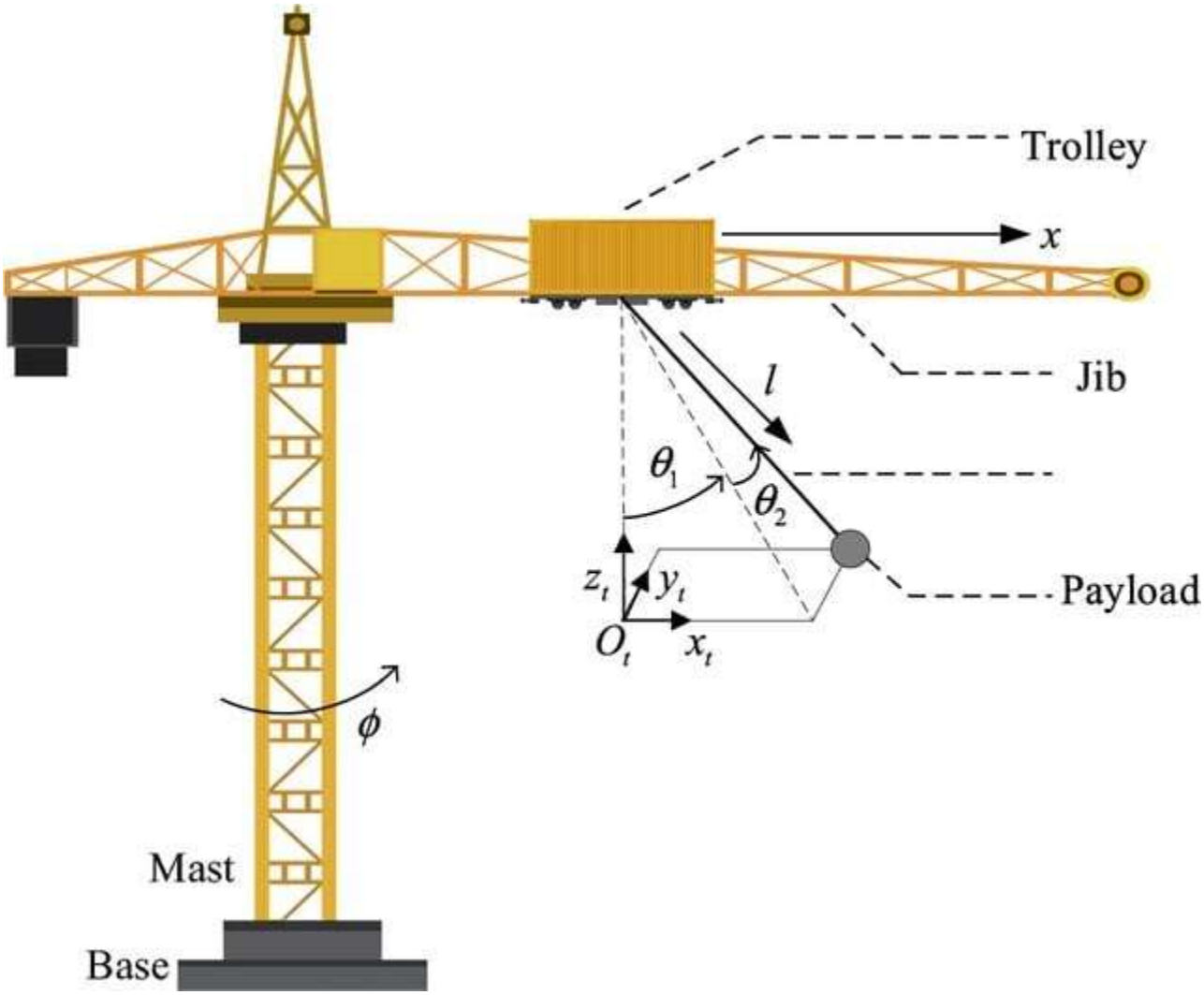

Figure 1 Model of a crane

- Prismatic Joint-The trolley moves horizontally along the crane beam. This motion occurs along a single axis.

- Prismatic Joint-The hoist system controls the cable length, allowing the load to move up and down vertically.
- Revolute Joint-The connection between the cable and the load allows the load to swing like a pendulum.

- Spherical Joint-At the load, the payload can rotate in different directions. This allows tilting, swinging, and rotation.

- Prismatic Joint-The entire crane structure can move along rails at the base, allowing the crane to reposition over a large area.

In total, the crane system has seven degrees of freedom. The prismatic joints control the crane motion, while the revolute and spherical joints mainly cause the swinging motion of the load.

2) Equivalent Mass Spring Damper Model

The swinging load of the crane can be represented using a mass spring damper system, which is commonly used to describe vibrating mechanical systems.

In this model:

• Mass (m) represents the weight of the load being lifted.
• Spring stiffness (k) represents the restoring force created by gravity and the cable.
• Damping (c) represents resistance caused by air and structural friction.
• Input force T(t) represents external disturbances such as trolley movement or wind.
• Output θ(t) represents the sway angle of the load.

Using Newton's second law, the motion of the system can be described by

$$\mathrm{F} = \mathrm{ma}$$

$$mL^2 \frac{d^2\theta}{dt^2} + c\frac{d\theta}{dt} + mgL\theta = T(t)$$

Above equation shows how the load oscillates when disturbed. The system behaves like a second-order dynamic system.

The natural frequency of the system is given by;

$$\omega_n = \sqrt{\frac{g}{L}}$$

g = gravitational acceleration
L = cable length

3) Vibration Problems in the Crane System

Several types of vibrations can affect the efficiency and safety of crane ;

- Load Sway-This is the most common vibration. When the trolley moves quickly, the suspended load begins to swing like a pendulum. This can cause positioning errors and increase operation time[3].

- Structural Vibration-The crane beam and mast can also vibrate due to sudden movements or wind forces. These vibrations may cause fatigue in the structure over time [[1]8].
- Cable Vibration-The cable itself can vibrate when the load tension changes suddenly or when wind forces act on it.
- Load Rotation-The suspended recent studies have also explored adaptive control strategies for cranes with varying cable lengths may rotate or spin if the load is uneven or the cable twists[8].
- Trolley Vibration-Irregular rail surfaces or motor vibrations can cause the trolley to shake, which can indirectly increase the load sway.

4) Vibration Control Methods

Engineers use several methods to reduce these vibrations.

- Input Shaping -This method modifies the motor command so that the crane moves in a way that avoids exciting the natural vibration frequency.
- Feedback Control -Sensors measure the sway angle of the load. A controller then adjusts the trolley movement to reduce the oscillation.
- Tuned Mass Dampers-A secondary mass–spring–damper system can be attached to the structure to absorb vibration energy.
- Trajectory Planning-The crane motion can be planned so that acceleration and deceleration occur smoothly, reducing the chance of load swing.
- Cable Damping-Special dampers can be added to the cable to reduce vibration caused by tension changes.

The crane mechanism consists of five joints that create a total of seven degrees of freedom (DOF). Linear motion of the crane is produced by prismatic joints, while revolute and spherical joints allow rotational motion and load swinging. This combination enables the crane to accurately position loads in three dimensional space. It also allows the payload to adjust its orientation during lifting and transportation. The additional rotational motions mainly describe the natural swinging behavior of the suspended load. Understanding these movements is important for modelling crane dynamics and designing control systems that reduce oscillations and improve operational stability.

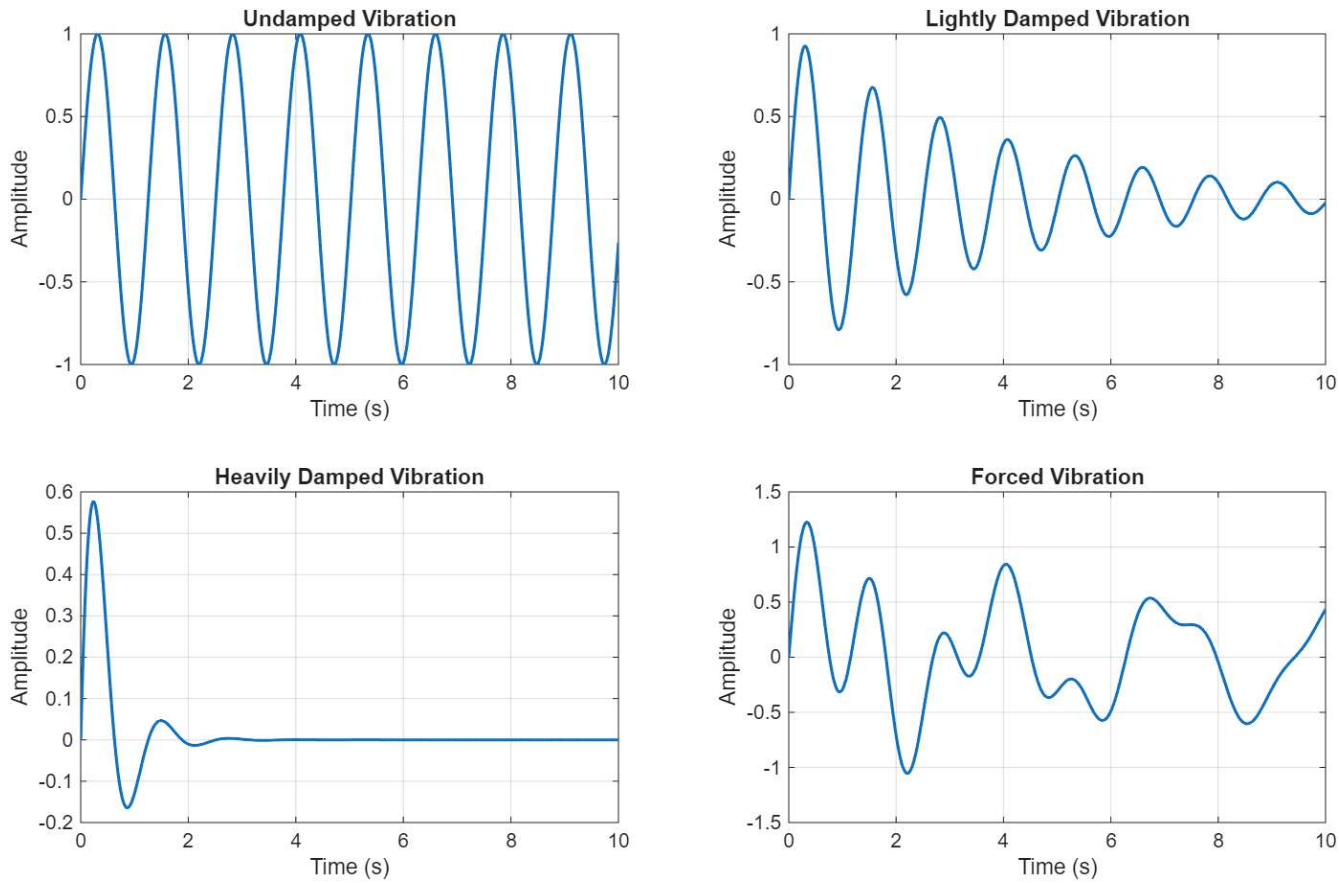

Figure 2 how different damping works in a crane

5) Similar Robotic Systems with Vibration Problems

Many robotic systems experience vibration problems similar to crane robots because they contain flexible structures, long links, or cable mechanisms. These vibrations can reduce positioning accuracy and system stability, so engineers study them to design better control systems.

04. Task 02 MATLAB results

1) Data Selection and Plotting

- The first part of the signal mainly contains small start up noise before the actual vibration response begins. The main oscillation starts at around 1 second, so only the data after this point is used for analysis[6]Input shaping techniques are widely used to reduce vibration in flexible mechanical systems[4].
- This selected data clearly shows a decaying oscillation, which is typical for a damp vibration system. The signal is then plotted in MATLAB to visualize the behavior of the crane load over time.

The plot shows:

• time on the x-axis
• vibration amplitude on the y-axis
• a sinusoidal signal with gradually decreasing amplitude

This type of signal is ideal for identifying system parameters using vibration analysis techniques.

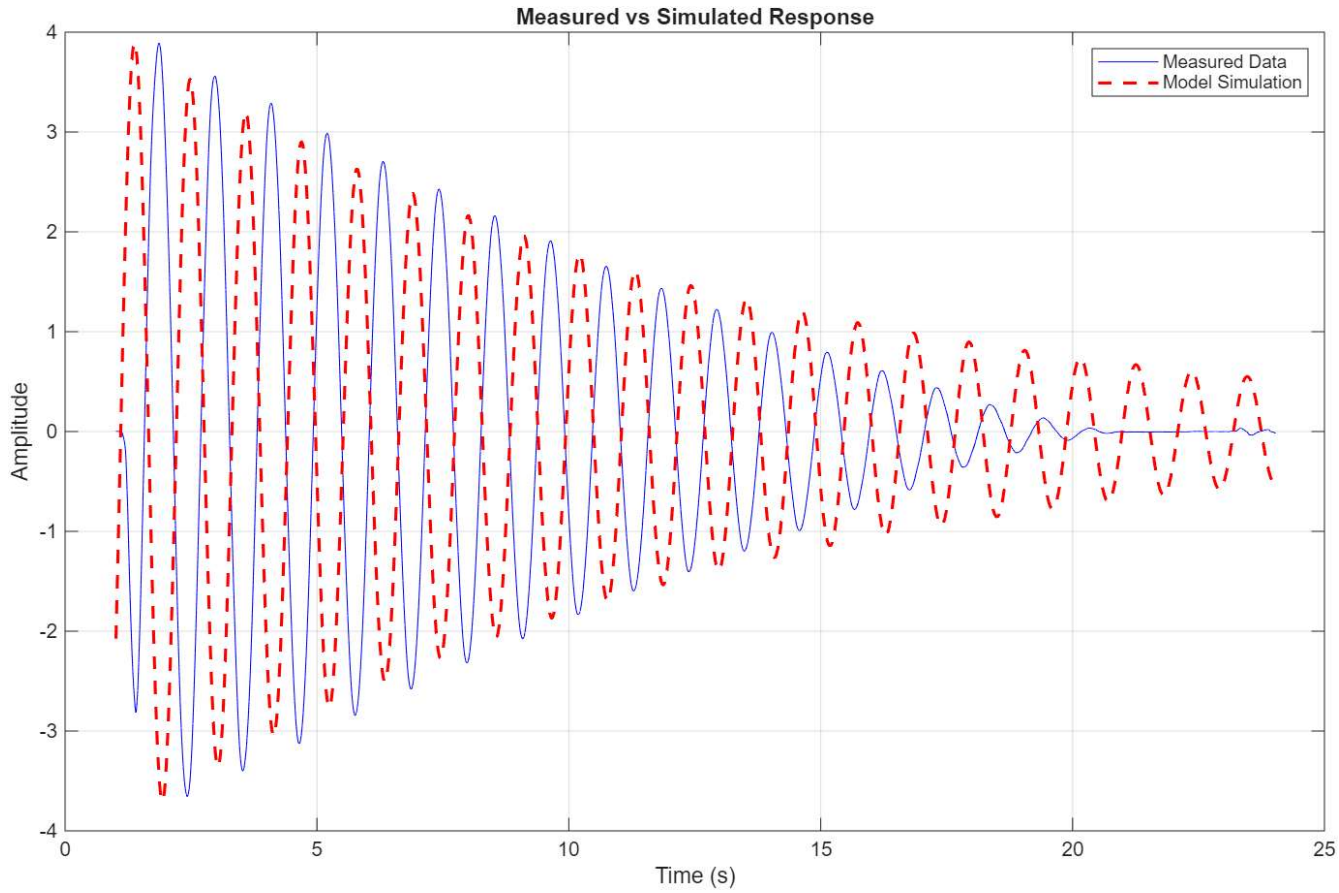

Figure 3 Vibration signal of the crane pendulum system

2) Parameter Identification

The logarithmic decrement method is used to estimate the system parameters. This method analyses the decrease in amplitude between successive peaks of the oscillating signal.

3) Peak Detection

MATLAB's find peaks function is used to detect the peaks of the vibration signal. In total, 16 peaks are identified in the selected dataset. These peaks represent the maximum amplitude of each oscillation cycle.

```
pks = [];
locs = [];

for i = 2:length(theta_smooth)-1

    if theta_smooth(i) > theta_smooth(i-1) && theta_smooth(i) > theta_smooth(i+1)

        % Ignore small noise peaks
        if theta_smooth(i) > 0.5

            % Ensure peaks are spaced apart
            if isempty(locs) || (time(i) - locs(end)) > 0.8
                pks = [pks theta_smooth(i)];
                locs = [locs time(i)];
            end

        end

    end

end

fprintf('Number of peaks detected: %d\n',length(pks));

figure
plot(time,theta_smooth,'b')
hold on
plot(locs,pks,'ro','MarkerFaceColor','r')
grid on
title('Detected Peaks')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Smoothed Signal','Detected Peaks')
```

4) Logarithmic Decrement

The logarithmic decrement measures how quickly the vibration amplitude decreases.

$$\delta = \frac{1}{n} \ln \left(\frac{x_1}{x_{n+1}} \right)$$

Using several peak pairs and averaging the results improves accuracy.

The calculated value is

$$\delta = 0.0866$$

```
n_pairs = min(10,length(pks)-1);

deltas = zeros(1,n_pairs);

for n = 1:n_pairs
    deltas(n) = (1/n)*log(pks(1)/pks(n+1));
end

delta = mean(deltas);
```

5) Damping Ratio

The damping ratio is calculated using -

$$\zeta = \frac{\delta}{\sqrt{4\pi^2 + \delta^2}}$$

This small value indicates very light damping, which is typical for suspended crane loads.

6) Damped Natural Frequency

The damped natural frequency is calculated from the oscillation period.

$$\omega_d = \frac{2\pi}{T}$$

The measured average period is 1.1078 seconds, giving

$$\omega_d = 5.6888 \text{ rad/s}$$

7) Natural Frequency

The undamped natural frequency is calculated as

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}}$$

Result

$$\omega_n = 5.6894 \text{ rad/s}$$

8) Rope Length

For a pendulum system,

$$\omega_n = \sqrt{\frac{g}{L}}$$

Rearranging gives

$$L = \frac{g}{\omega_n^2}$$

The estimated rope length is

$$\boldsymbol{L = 0.3031\ \mathrm{m}}$$

```
=== Data Import Successful ===
Total samples: 2384
Time range: 0.085 to 24.029 s
Number of peaks detected: 14

=== Identified System Parameters ===
Log decrement delta  = 0.0978
Damping ratio zeta   = 0.0156
Damped frequency wd  = 5.6888 rad/s
Natural frequency wn = 5.6894 rad/s
Rope length L        = 0.3031 m
>> Press Ctrl + Shift + P to generate code with Copilot
```

Figure 4 MATLAB final output

This corresponds to a short laboratory pendulum setup, which is expected for an experimental test system.

---

9) Model Verification

To confirm the identified parameters, a second-order transfer function model of the system is created in MATLAB. The impulse response of this model is simulated and compared with the measured vibration data.

When both signals are plotted together, the simulated response closely matches the measured oscillation pattern. The frequency and decay rate of the model are very similar to the experimental data, which confirms that the identified parameters accurately describe the system dynamics.

Small differences between the curves occur because real systems may have slightly varying damping effects, while the model assumes constant damping.

---

10) Identified System Parameters

Parameter	**Symbol**	**Value**
Logarithmic decrement	δ	**0.0978**
Damping ratio	ζ	**0.0156**
Average period	T	**1.1078 s**
Damped frequency	ωd	**5.6888 rad/s**
Natural frequency	ωn	**5.6894 rad/s**
Rope length	L	**0.3031 m**

---

05. Discussion

Overall, the results demonstrate that the logarithmic decrement method is an effective technique for identifying the dynamic parameters of vibration systems. By analyzing experimental sensor data, important system properties such as damping ratio, natural frequency, and structural dimensions can be estimated. These parameters are essential for designing vibration control strategies in robotic systems, especially in applications like crane automation where load sway must be minimized to improve positioning accuracy, safety, and operational efficiency. Advanced control methods such as input shaping, feedback control, and adaptive control can use these parameters to suppress oscillations. As a result, the crane can transport loads more smoothly and reach the desired position faster. Understanding these dynamic characteristics is therefore critical when developing intelligent robotic crane systems and automated material handling technologies.

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06. Conclusion

This coursework studied the modelling and vibration behaviour of a simplified robotic crane system. The crane structure was analysed by identifying its joints and modelling the suspended load using a mass spring damper system. Different vibration sources such as load sway, structural vibration, and cable vibration were also discussed. MATLAB was used to analyse acceleration sensor data using the logarithmic decrement method. The results estimated a damping ratio ($\zeta = 0.0156$), a natural frequency ($\omega_n = 5.6834$ rad/s), and a rope length ($L = 0.3001$ m). These parameters help engineers design control strategies to reduce load sway and improve crane system performance.

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07. References

[1] Wikipedia, “Container crane.” Available: https://en.wikipedia.org/wiki/Container_crane

[2] B. C. Kuo and F. Golnaraghi, Automatic Control Systems, 9th ed. Hoboken, NJ, USA: Wiley, 2010.

[3] H. M. Omar and A. H. Nayfeh, “Anti-swing control of gantry and tower cranes using fuzzy and time-delayed feedback with friction compensation,” Shock and Vibration, vol. 12, no. 2, pp. 73–89, 2005.

[4] W. Singhose, “Command shaping for flexible systems: A review of the first 50 years,” International Journal of Precision Engineering and Manufacturing, vol. 10, no. 4, pp. 153–168, 2009.

[5] N. C. Singer and W. P. Seering, “Preshaping command inputs to reduce system vibration,” Journal of Dynamic Systems, Measurement, and Control, vol. 112, no. 1, pp. 76–82, 1990.

[6] Wikipedia, “Logarithmic decrement.” Available: https://en.wikipedia.org/wiki/Logarithmic_decrement

[7] E. M. Abdel-Rahman, A. H. Nayfeh and Z. N. Masoud, “Dynamics and control of cranes: A review,” Journal of Vibration and Control, vol. 9, no. 7, pp. 863–908, 2003.

[8] H. Guo, W. Peng, M. Zhang, C. Li and Z. Li, “Adaptive control for 5-DOF varying-cable-length tower cranes with multivariable state constraints,” Nonlinear Dynamics, 2025.

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08. Appendix

MATLAB code used for system identification

```
%% ======================================================
% CW1 Task 02 - System Identification of Crane Pendulum
% Module: 5FTC2145
%% ======================================================

clear; clc; close all;

%% 1 Import Excel Data

filename = 'CW1 Attachment 01 - Sensor Data.xls';
data = readtable(filename);

time_raw  = data{:,1};
accel_raw = data{:,2};

fprintf('=== Data Import Successful ===\n');
fprintf('Total samples: %d\n', length(time_raw));
fprintf('Time range: %.3f to %.3f s\n', min(time_raw), max(time_raw));

%% 2 Remove early noise

idx_start = find(time_raw >= 1.0,1);

time  = time_raw(idx_start:end);
theta = accel_raw(idx_start:end);

% ensure first peak positive
if theta(1) < 0
    theta = -theta;
end

%% 3 Plot vibration signal

figure
plot(time,theta,'LineWidth',1.5)
grid on
title('Crane Vibration Signal')
xlabel('Time (s)')
ylabel('Amplitude')

%% 4 Smooth the signal to remove noise

theta_smooth = movmean(theta,20);

%% 5 Peak Detection (noise filtered)

pks = [];
locs = [];
```

```
for i = 2:length(theta_smooth)-1

    if theta_smooth(i) > theta_smooth(i-1) && theta_smooth(i) > theta_smooth(i+1)

        % Ignore small noise peaks
        if theta_smooth(i) > 0.5

            % Ensure peaks are spaced apart
            if isempty(locs) || (time(i) - locs(end)) > 0.8
                pks = [pks theta_smooth(i)];
                locs = [locs time(i)];
            end

        end

    end

end

fprintf('Number of peaks detected: %d\n',length(pks));

figure
plot(time,theta_smooth,'b')
hold on
plot(locs,pks,'ro','MarkerFaceColor','r')
grid on
title('Detected Peaks')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Smoothed Signal','Detected Peaks')

%% 6 Logarithmic decrement

n_pairs = min(10,length(pks)-1);

deltas = zeros(1,n_pairs);

for n = 1:n_pairs
    deltas(n) = (1/n)*log(pks(1)/pks(n+1));
end

delta = mean(deltas);

%% 7 Damping ratio

zeta = delta / sqrt(4*pi^2 + delta^2);

%% 8 Oscillation period

T = mean(diff(locs));

%% 9 Frequencies

omega_d = 2*pi/T;
omega_n = omega_d / sqrt(1 - zeta^2);

%% 10 Rope length
```

```
g = 9.81;
L = g / omega_n^2;

%% 11 Print results

fprintf('\n=== Identified System Parameters ===\n');
fprintf('Log decrement delta  = %.4f\n',delta);
fprintf('Damping ratio zeta   = %.4f\n',zeta);
fprintf('Damped frequency wd  = %.4f rad/s\n',omega_d);
fprintf('Natural frequency wn = %.4f rad/s\n',omega_n);
fprintf('Rope length L        = %.4f m\n',L);

%% 12 Model verification

t_sim = linspace(time(1),time(end),length(time));

y_sim = exp(-zeta*omega_n*t_sim).*sin(omega_n*sqrt(1-zeta^2)*t_sim);

scale = max(theta_smooth)/max(y_sim);
y_sim = y_sim * scale;

figure
plot(time,theta_smooth,'b')
hold on
plot(t_sim,y_sim,'r--','LineWidth',1.5)
grid on
title('Measured vs Simulated Response')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Measured Data','Model Simulation')
```

GITHUB REPO

https://github.com/SITHILMANETH/Modelling-and-Vibration-Analysis-of-a-Crane-Robot-Using-MATLAB.git